

Lab 11

Program 1

A **class** is a blueprint for creating objects. It contains properties (variables) and methods (functions).

Syntax:

```
class ClassName {  
    constructor(parameter1, parameter2) {  
        this.parameter1 = parameter1;  
        this.parameter2 = parameter2;  
    }  
  
    methodName() {  
        // code  
    }  
}  
  
let obj = new ClassName(value1, value2);  
obj.methodName();
```

Question:

Create a **Car** class containing:

- company
- model

Display car details using method **display()**

Program 2

Spread operator (...) expands an array or object into individual elements.

It is mainly used for:

- Copying arrays

```
let copiedArray = [...arr1];
```

- Merging arrays

```
let combinedArray = [...arr1, ...arr2];
```

- Copying objects
 - `Let copiedObject = {...obj1}`
- Merging objects

```
let combinedObject = { ...obj1, ...obj2 };
```

Task to perform:

- ➔ Make a student object named StudentObj with properties : Name , Age
- ➔ Make roll object RollObj with properties: Roll no. , SPI
- ➔ Merge these two objects using spread operator.

Program 3

3. Demonstrate the 'for of' loop. (A)

```
let marks = [78, 85, 92, 88, 76];
```

//Syntax

```
<!--
for(variableName of iterableArrayName)
{
    // statements
}
-->
```

Task:

Calculate the total marks and Average marks in marks array.

Program 4

4. Demonstrate the Array and Object Destructuring. (B)

Used to unpack Array(or Object) and assign the element to another variables

For example:

```
let arr = [1, 2, 3, 4, 5];
let [a, b, c, ...RestArray] = arr;
```

Therefore to get first element of Array as `arr[0]` → 1,

We can now get it as `console.log(a)` → 1

```
let obj = { name: "Peter", age: 24, city: "Queens" };
let { name, age } = obj; //destructuring pf object.
document.write("name: " + name + "<br>");
document.write("age: " + age + "<br>");
```

Object Destructuring : Object destructuring extracts properties from an object and assigns them to variables of the same name

So now instead of writing `console.log(obj.name) → Peter`

You can now write `console.log(name) → Peter` (Only allowed when the object is destructured)